

FINAL PROJECT #7

Course: CHE 433. PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2019
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Team Names: _____
Due Dates: Oral Presentation Monday December 09, and
Tuesday December 10 in my office: EIB 244

Total Number of points.....100
Project Presentation (max 80 points).....
Oral Examination (max 20 points).....

Note: All problems are worth 100 points. Your final score will be the average value

Problem 7A: Propane Dehydrogenation Technology Project

Prepare a detailed process design of a world class propane dehydrogenation manufacturing plant located in the US Gulf coast for the production of polymer grade (PG) propylene (minimum 99.5 wt% purity) using propane as feedstock. The nominal capacity of the plant is 450 KMTA with a 90% operating factor. Your final report should contain the following sections:

- Heat and Weight Balance
- Sizing and Total Fixed Capital Cost of the major equipment using the standard cost estimating Excel spreadsheet
- Operating Costs including utilities and raw material

Milestones for successful completion of the project

- Milestone 1: Literature research for the kinetic mechanism of propane dehydrogenation
- Milestone 2: Preparation of the block flow diagram of the dehydrogenation technology
- Milestone 3: Incorporation of the kinetic model and rigorous distillation columns
- Milestone 4: Estimation of fixed capital cost and cost of utilities
- Milestone 5: Pinch Analysis and Energy Optimization

For Operating Cost Calculation use the following Standard Prices for 2010.

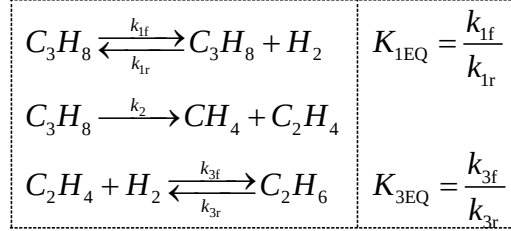
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|-------------------|-------------------|
| 1. Electricity: | 0.09 USD/kWh |
| 2. HP Steam: | 11.45 USD/1000lb |
| 3. MP Steam: | 9.50 USD/1000lb |
| 4. LP Steam: | 8.48 USD/1000lb |
| 5. Cooling Water: | 0.08 USD/1000gal |
| 6. Fuel Gas(LHV) | 7.30 USD/MMBtu |
| 7. Propane Price: | 700.00 USD/1000kg |

PROPANE FEEDSTOCK DATA

Stream Conditions	Propane Feedstock
Hydrogen, %wt	0.00
Methane, %wt	0.00
Ethylene, %wt	0.00
Ethane, %wt	2.00
Propylene, %wt	0.00
Propane, %wt	95.0
n-Butane, %wt	3.00
Total Flow, kg/hr	
Temperature, deg C	30.0
Pressure, barg	25.0

Kinetic Model for Propane Dehydrogenation

The following kinetic model has been implemented in AthenaVisual Studio based on the paper “*Transient kinetic modeling of propane dehydrogenation over a Pt–Sn–K/Al₂O₃ catalyst*” by M.P. Lobera, C. Tellez, J. Herguido and M. Menendez, Aragon Institute for Engineering Research (I3A), University of Zaragoza, 50009 Zaragoza, Spain.



The reaction rates are given by the following equations:

$\mathfrak{R}_1 = \frac{k_{1f} \left(P_{C_3H_8} - \frac{P_{H_2} P_{C_3H_6}}{K_{1EQ}} \right)}{1 + K_A P_{C_3H_6}}$	$k_{1f} = k_{1fo} \exp \left(-\frac{E_{1a}}{R} \left(\frac{1}{T} - \frac{1}{T_o} \right) \right)$	$K_A = K_{Ao} \exp \left(-\frac{\Delta H_A}{R} \left(\frac{1}{T} - \frac{1}{T_o} \right) \right)$
$\mathfrak{R}_2 = k_2 P_{C_3H_8}$	$k_2 = k_{2o} \exp \left(-\frac{E_{2a}}{R} \left(\frac{1}{T} - \frac{1}{T_o} \right) \right)$	$\ln K_A = \ln K_{Ao} + \frac{\Delta H_A}{RT_o} - \frac{\Delta H_A}{RT}$
$\mathfrak{R}_3 = k_{3f} \left(P_{C_2H_4} P_{H_2} - \frac{P_{C_2H_6}}{K_{3EQ}} \right)$	$k_{3f} = k_{3fo} \exp \left(-\frac{E_{3a}}{R} \left(\frac{1}{T} - \frac{1}{T_o} \right) \right)$	

The values of the kinetic and adsorption constants are given in the table below:

Constant	Value	Constant	Value
k_{1fo}	0.03145200 kmol ⁻¹ kg ⁻¹ hr ⁻¹ bar ⁻¹	E_{1a}	34.57 kJ mol ⁻¹
k_{2fo}	0.00027900 kmol ⁻¹ kg ⁻¹ hr ⁻¹ bar ⁻¹	E_{2a}	137.31 kJ mol ⁻¹
k_{3fo}	0.00001416 kmol ⁻¹ kg ⁻¹ hr ⁻¹ bar ⁻²	E_{3a}	154.54 kJ mol ⁻¹
K_{Ao}	3.46 bar ⁻¹	ΔH_A	-85.817 kJ mol ⁻¹
T_o	520 °C	$\ln K_{Ao} + \frac{\Delta H_A}{RT_o}$	-11.772647
R	8.314 J mol ⁻¹ K ⁻¹	$\frac{\Delta H_A}{R}$	-10321.9870

The Equilibrium constants can be calculated from Gibbs free energy data and van't Hoff's equation.

ASPEN PLUS IMAGES FOR THE KINETIC MODEL IMPLEMENTATION

Configuration Kinetic Equilibrium Activity GLHHW Adsorption Information

No.	Name	Reaction class	Status	Reversible	Stoichiometry
1	RXN1	LHHW	On	☒	PROPANE(MIXED) --> PROPYLEN(MIXED) + HYDROGEN(MIXED)
2	RXN2	POWERLAW	On	☐	PROPANE(MIXED) --> METHANE(MIXED) + ETHYLENE(MIXED)
3	RXN3	POWERLAW	On	☐	ETHYLENE(MIXED) + HYDROGEN(MIXED) --> ETHANE(MIXED)

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Configuration Kinetic Equilibrium Activity GLHHW Adsorption Information

1) PROPANE(MIXED) --> PROPYLEN(MIXED) + HYDROGEN(MIXED)

Reaction class: LHHW Reacting phase: Vapor

[C_i] basis: Partial pressure

[C_i] units: bar

Rate basis: Cat (wt)

Rate units: kmol/kg-hr

LHHW kinetic expression

$r = \frac{[\text{Kinetic factor}][\text{Driving force}]}{[\text{Adsorption}]}$

Kinetic factor

If T₀ is specified: Kinetic factor = $k(T/T_0)^n e^{-(E/R)[1/T-1/T_0]}$

If T₀ is not specified: Kinetic factor = $kT^n e^{-E/RT}$

k: 0.031452

n: 0

E: 34.57 kJ/mol

T₀: 520 C

Solids Driving Force Adsorption

Specifications Summary

Adsorption Expression

Ads.=[Term1+Term2+...]^Ads. Expression exponent; Termi = Ki*[C1]^n1*[C2]^n2 ...

Reacting phase: Vapor

[C] basis: Partial pressure

Adsorption expression exponent: 1

Component	Term no. 1	Term no. 2	Term no. 3
PROPYLEN	0	1	

Ads.=[K1 + K2*[PROPYLEN]]

Adsorption constants

Term No.: 1

A: 0 B: 0 C: 0 D: 0

K1=1

Adsorption Expression

Ads.=[Term1+Term2+...]^Ads. Expression exponent; Termi = Ki*[C1]^n1*[C2]^n2 ...

Reacting phase: Vapor

[C] basis: Partial pressure

Adsorption expression exponent: 1

Component	Term no. 1	Term no. 2	Term no. 3
PROPYLEN	0	1	

Ads.=[K1 + K2*[PROPYLEN]]

Adsorption constants

Term No.: 2

A: -11.7726 B: -10322 C: 0 D: 0

K2=7.7E-06*exp(-10322/T)

um
Activity
GLHH

2) PROPANE(MIXED) --> METHANE(MIXED) + ETHYLENE(MIXED)

Reaction class: POWERLAW

Reacting phase: Vapor

[Ci] basis: Partial pressure

[Ci] units: bar

Rate basis: Cat (wt)

Rate units: kmol/kg-hr

Powerlaw kinetic expression

r = [Kinetic factor][Driving force]

Kinetic factor

If To is specified: Kinetic factor = $k(T/T_o)^n e^{-(E/R)[1/T-1/T_o]}$

If To is not specified: Kinetic factor = $kT^n e^{-E/RT}$

k: 0.000279

n: 0

E: 137.31 kJ/mol

To: 520 C

Solids

Driving Force

Adsorption

Specifications

Summary

Configuration

Kinetic

Equilibrium

Activity

GLHHW Adsorption

Information

3) ETHYLENE(MIXED) + HYDROGEN(MIXED) --> ETHANE(MIXED)

Reaction class: POWERLAW

Reacting phase: Vapor

[Ci] basis: Partial pressure

[Ci] units: bar

Rate basis: Cat (wt)

Rate units: kmol/kg-hr

Powerlaw kinetic expression

r = [Kinetic factor][Driving force]

Kinetic factor

If To is specified: Kinetic factor = $k(T/T_o)^n e^{-(E/R)[1/T-1/T_o]}$

If To is not specified: Kinetic factor = $kT^n e^{-E/RT}$

k: 1.416e-05

n: 0

E: 154.54 kJ/mol

To: 520 C

Solids

Driving Force

Adsorption

Specifications

Summary

Catalyst volume per reactor	31m ³
Reactor Bulk Density	650 kg / m ³
Reactor Bed Porosity	0.42
Catalyst Mass Density	1200 kg / m ³
Catalyst Mass per Reactor	20150 kg
Catalyst Diameter	1.8 mm
WHSV, hr⁻¹	
Propane Conversion per Pass	33.6%wt
Propane Selectivity per Pass	89.2%wt
Reactor Length	10 m
Reactor Outside Diameter	4 m
Annulus Width	0.5 m

Process Simulation with ASPEN Plus

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TEST FEED FOR KINETIC MODEL TUNING

Stream Conditions	Feed	Product
Hydrogen, %wt	6.10	7.20
Methane, %wt	1.30	2.30
Ethylene, %wt	0.00	0.80
Ethane, %wt	0.80	1.20
Propylene, %wt	0.20	27.4
Propane, %wt	90.9	60.4
n-Butane, %wt	0.70	0.70
Total Flow, kg/hr	190,330	190,330
Temperature, deg C	650.0	
Pressure, bar	~ 2	
Catalyst Mass per Reactor, kg	20,150	
Number of Reactors in Series	4	

COLD BOX SPECIFICATIONS

Stream Conditions	Vapor	Liquid
Hydrogen, split fraction	1.000	0.000
Methane, split fraction	0.882	0.118
Ethylene, split fraction	0.095	0.905
Ethane, split fraction	0.095	0.905
Propylene, split fraction	0.000	1.000
Propane, split fraction	0.000	1.000
n-Butane, split fraction	0.000	1.000
Pressure, barg	14.0	14.0